EIGRP Questions

4 questions | Original English questions and source answers

Question 1

Refer to the exhibit.
Which route type does the routing protocol Code D represent in the output?

```
      10.0.0.0/24 is subnetted, 1 subnets
C        10.0.0.0 is directly connected, FastEthernet0/1
C     172.16.0.0/16 is directly connected, FastEthernet0/0
D     192.168.0.0/24 [90/30720] via 172.16.0.2, 00:00:04, FastEthernet0/0
```

Options

A. internal BGP route

B. statically assigned route

C. /24 route of a locally configured IP

D. route learned through EIGRP

Answer

D. route learned through EIGRP

Source explanation

Code “D” represents for EIGRP. Code “E” could not be used for EIGRP as it was used for Exterior Gateway Protocol (EGP), which is BGP predecessor.

Source: CCNA 200-301 Questions.html | Question 1

Question 2

A router running EIGRP has learned the same route from two different paths. Which parameter does the router use to select the best path?

Options

A. administrative distance

B. metric

C. as-path

D. cost

Answer

B. metric

Source: CCNA 200-301 Questions.html | Question 2

Question 3

By default, how does EIGRP determine the metric of a route for the routing table?

Options

A. It uses a default metric of 10 for all routes that are learned by the router

B. It counts the number of hops between the receiving and destination routers and uses that value as the metric

C. It uses the bandwidth and delay values of the path to calculate the route metric

D. It uses a reference Bandwidth and the actual bandwidth of the connected link to calculate the route metric

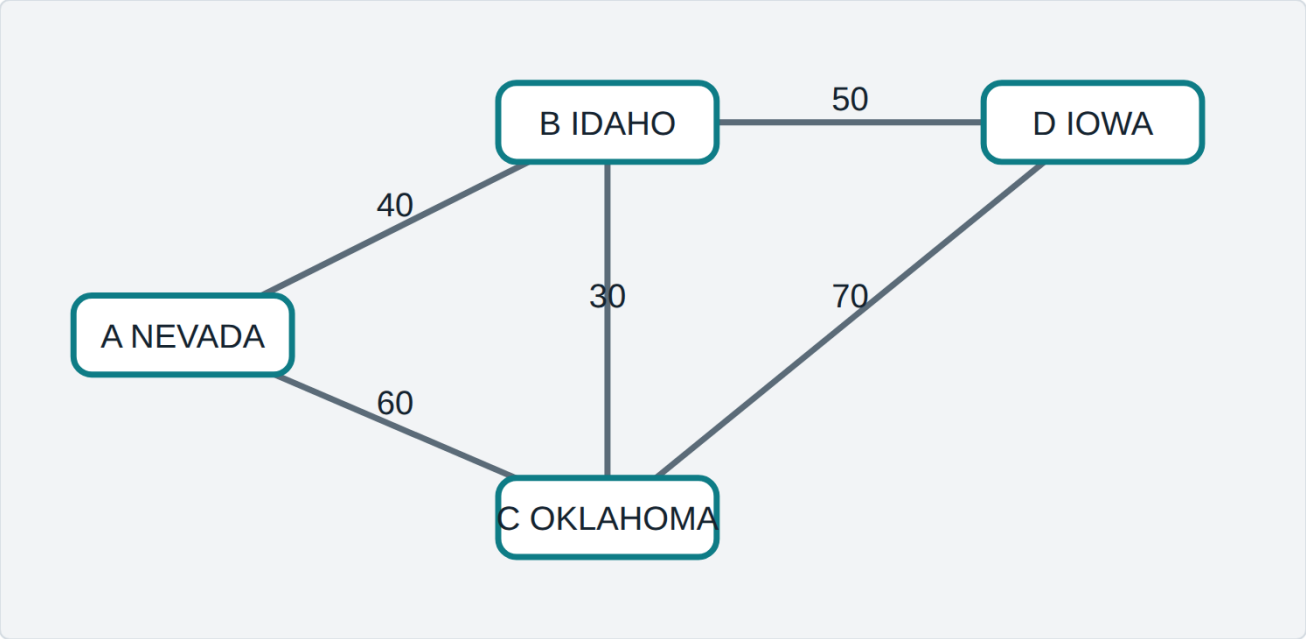
Answer

C. It uses the bandwidth and delay values of the path to calculate the route metric

Source: CCNA 200-301 Questions.html | Question 3

Question 4

Which two actions influence the EIGRP route selection process? (Choose two)



Options

A. The router calculates the feasible distance of all paths to the destination route

B. The advertised distance is calculated by a downstream neighbor to inform the local router of the bandwidth on the link

C. The router calculates the reported distance by multiplying the delay on the exiting interface by 256

D. The router calculates the best backup path to the destination route and assigns it as the feasible successor

E. The router must use the advertised distance as the metric for any given route

Answer

A. The router calculates the feasible distance of all paths to the destination route

D. The router calculates the best backup path to the destination route and assigns it as the feasible successor

Source explanation

The reported distance (or advertised distance) is the cost from the neighbor to the destination. It is calculated from the router advertising the route to the network. For example in the topology below, suppose routerA & answer 'The router calculates the best backup path to the destination route and assigns it as the feasible successor' are exchanging their routing tables for the first time. Router answer 'The router calculates the best backup path to the destination route and assigns it as the feasible successor' says “Hey, the best metric (cost) from me to IOWA is 50 and the metric from you to IOWA is 90” and advertises it to routerA. Router answer 'The router calculates the reported distance by multiplying the delay on the exiting interface by 256' considers the first metric (50) as the Advertised distance. The second metric (90), which is from NEVADA to IOWA (through IDAHO), is called the Feasible distance.

The reported distance is calculated in the same way of calculating the metric. By default (K1 = 1, K2 = 0, K3 = 1, K4 = 0, K5 = 0), the metric is calculated as follows:

metric = [10,000,000 / slowest bandwidth [in kbps] + sum of delay [in µsec] / 10] * 256

-> Answer 'The router calculates the reported distance by multiplying the delay on the exiting interface by 256' is not correct.

Feasible successor is the backup route. To be a feasible successor, the route must have an Advertised distance (AD) less than the Feasible distance (FD) of the current successor route -> Answer 'The router calculates the best backup path to the destination route and assigns it as the feasible successor' is correct.

Feasible distance (FD): The sum of the AD plus the cost between the local router and the next-hop router. The router must calculate the FD of all paths to choose the best path to put into the routing table.

Note: Although the new CCNA exam does not have EIGRP topic but you should learn the basic knowledge of this routing protocol.

Source: CCNA 200-301 Questions.html | Question 4